Quantified Self:

Performance Optimization Study

Analyzing 6 Weeks of Performance Metrics

Project Objective

Goal: Optimize work schedule based on empirical performance data

Challenge:

- Multiple concurrent priorities across different domains
- Academic coursework (Python, SQL, Statistics, Power BI)
- Content development (lesson plans, curricula)
- Professional development tasks
- Personal projects

Question: How can schedule design maximize output quality

while managing capacity effectively?

Methodology

Data Collection & Analysis Framework

Data Collection Method

Tracking Period: October 31 - November 26, 2025 (6 weeks)

Data Sources:

- AI-assisted activity logging system
- Timestamped work session records
- Output volume measurement per session

Variables Measured:

- Duration (hours per session)
- Output quality (1-10 scale, post-session evaluation)
- Activity category and complexity level
- Time of day (morning/afternoon/evening)
- Tasks initiated and completed

Performance Metrics: Calculation Methods

Energy (Objective)

```
Energy = (Output Quality × Task Complexity) ÷ 10
```

Productivity Score

```
Productivity = (Hours × Energy × Quality) ÷ 100
```

Completion Rate

```
Completion = (Tasks Completed ÷ Tasks Started) × 100
```

Task Efficiency

```
Efficiency = Tasks Completed ÷ Hours Worked
```

Task Complexity Scoring System

Complexity scores assigned by cognitive demand and task type:

High Complexity (10/10):

- Python programming, Database development
- Abstract reasoning, working memory
- Problem decomposition required

Medium-High (8-9/10):

- Privacy analysis, Advanced statistics
- Power BI, Business intelligence
- Sustained focus required

Medium (6-7/10):

- Business development, Strategy
- ESL curriculum, Lesson planning
- Moderate cognitive load

Low-Medium (4-5/10):

- Technical troubleshooting
- Vehicle research, Health topics
- Routine cognitive tasks

Low Complexity (2-3/10):

- Fitness planning, Music practice
- Gaming, Casual creative work
- Minimal cognitive demand

Scoring Basis:

Based on established cognitive load research and task characteristics

Dataset Summary: 6-Week Period

November 1-26, 2024 | Comprehensive productivity tracking

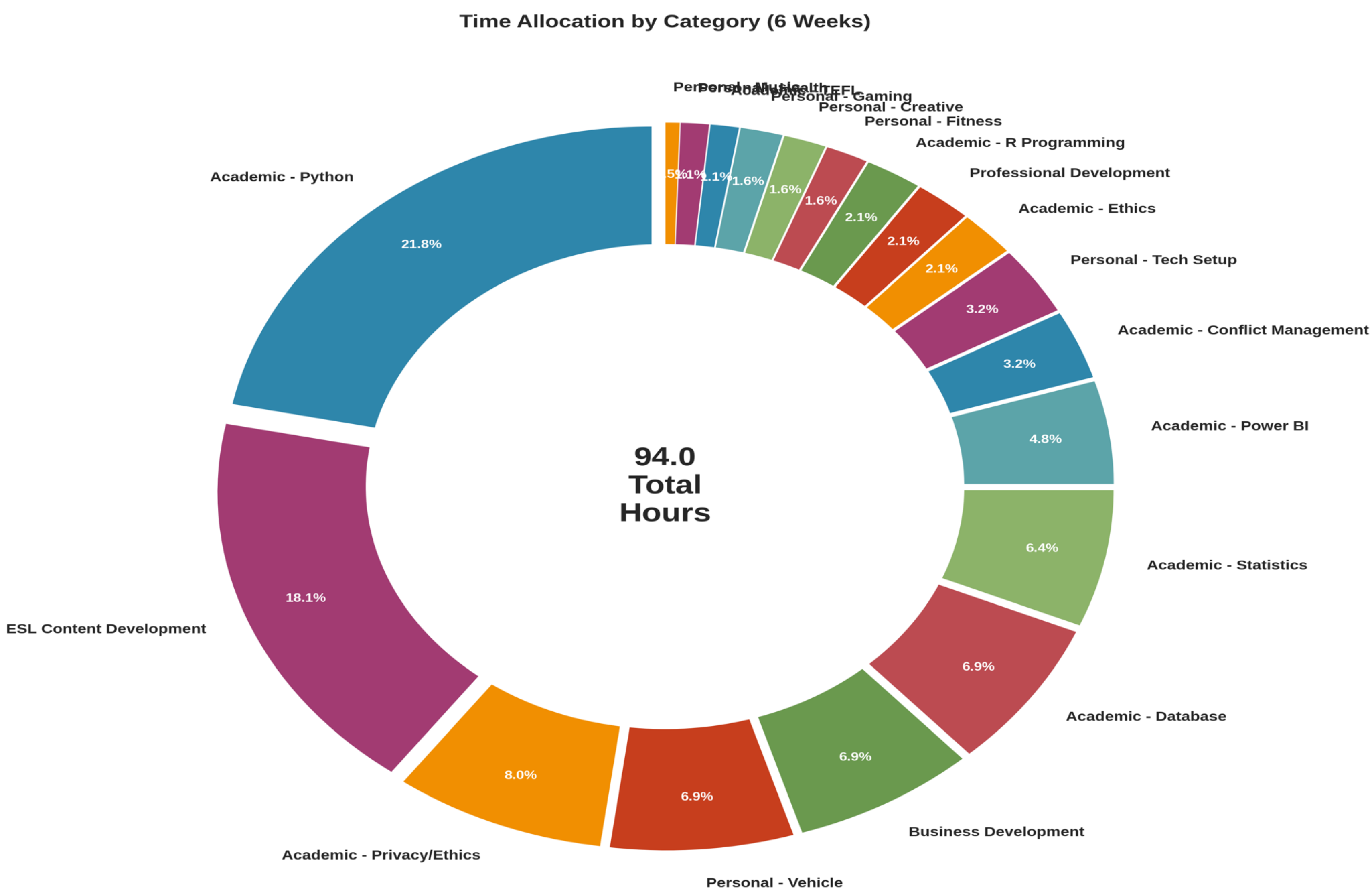
94 hrs Total Hours	**57** Work Sessions	**78 of 91** Tasks Completed
5.9/10 Avg Energy	**8.3/10** Avg Quality	**85.7%** Completion Rate

Data collected through timestamped work session records with quality assessments and task complexity ratings for each activity

Findings

Data Analysis Results

Finding 1: Time Distribution Across Categories



Finding 2: Two Types of Productive Capacity

Focus Capacity (Cognitive) vs. Output Capacity (Volume)

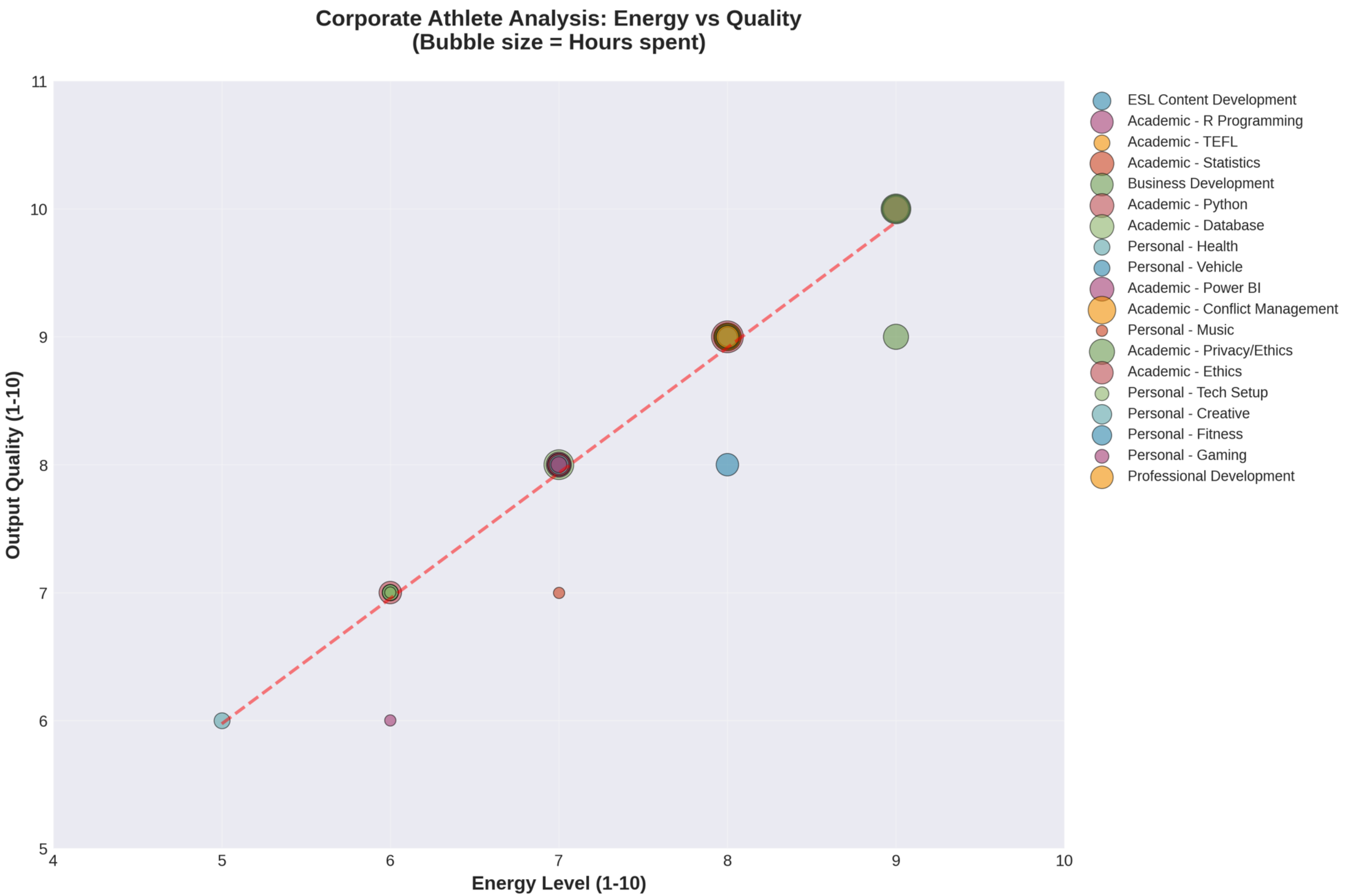
Morning (8am-12pm):
- Energy: 7.9/10 (Focus Capacity)
- Quality: 8.6/10
- Tasks: 20 (Output Capacity: 3.4/10)
- Avg Complexity: 9.3/10

Afternoon (12pm-5pm):
- Energy: 6.3/10 (Focus Capacity)
- Quality: 8.2/10
- Tasks: 20 (Output Capacity: 3.6/10)
- Avg Complexity: 7.7/10

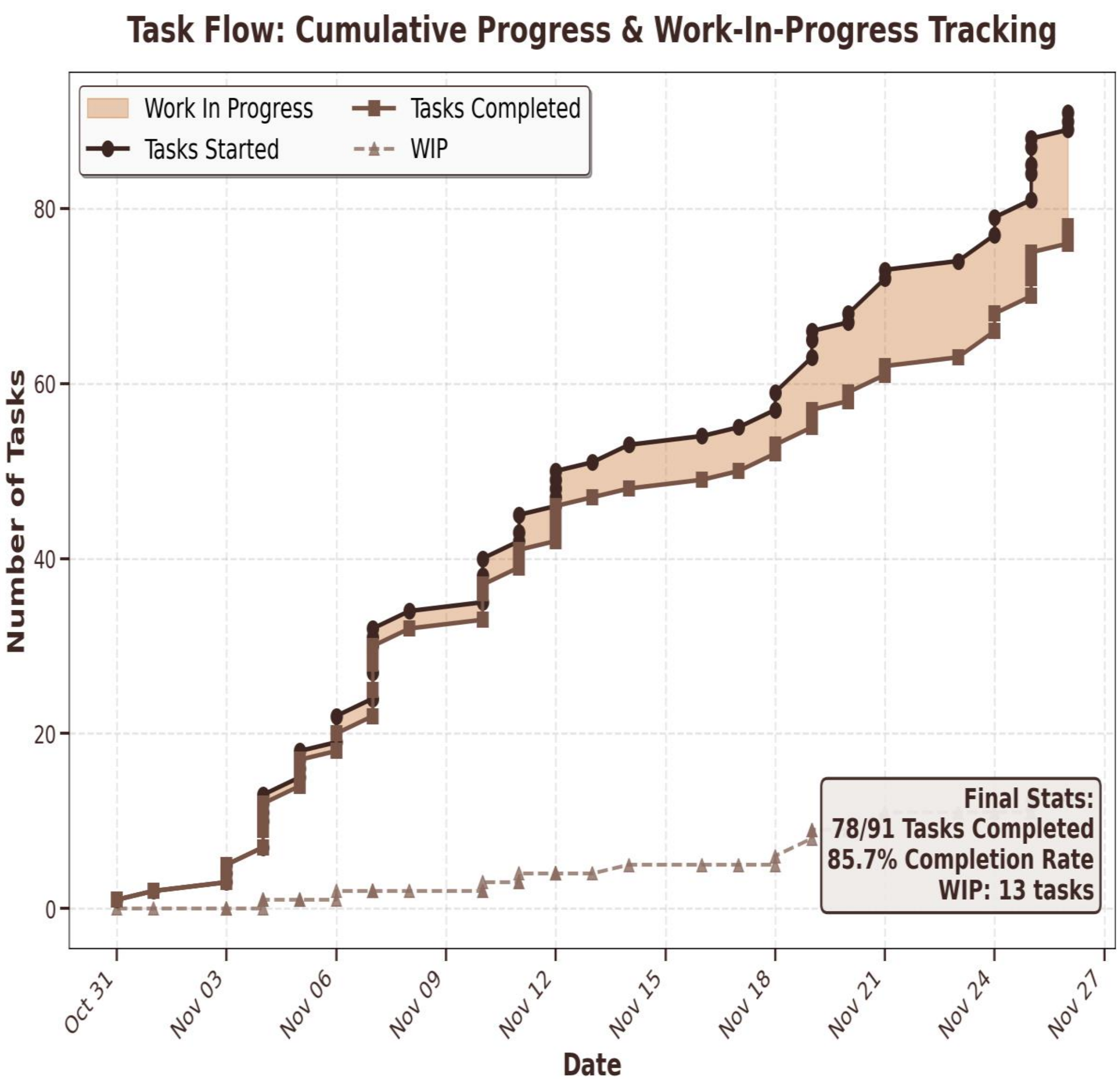
Evening (5pm-10pm):
- Energy: 3.7/10 (Focus Capacity)
- Quality: 7.9/10
- Tasks: 38 (Output Capacity: 7.5/10)
- Avg Complexity: 4.8/10

Key Insight: Different times excel at different types of work. Morning = highest focus capacity for complex analytical tasks. Evening = highest output capacity for high-volume simple tasks.

Finding 3: Energy Reflects Cognitive Deployment



Finding 4: Task Completion Performance

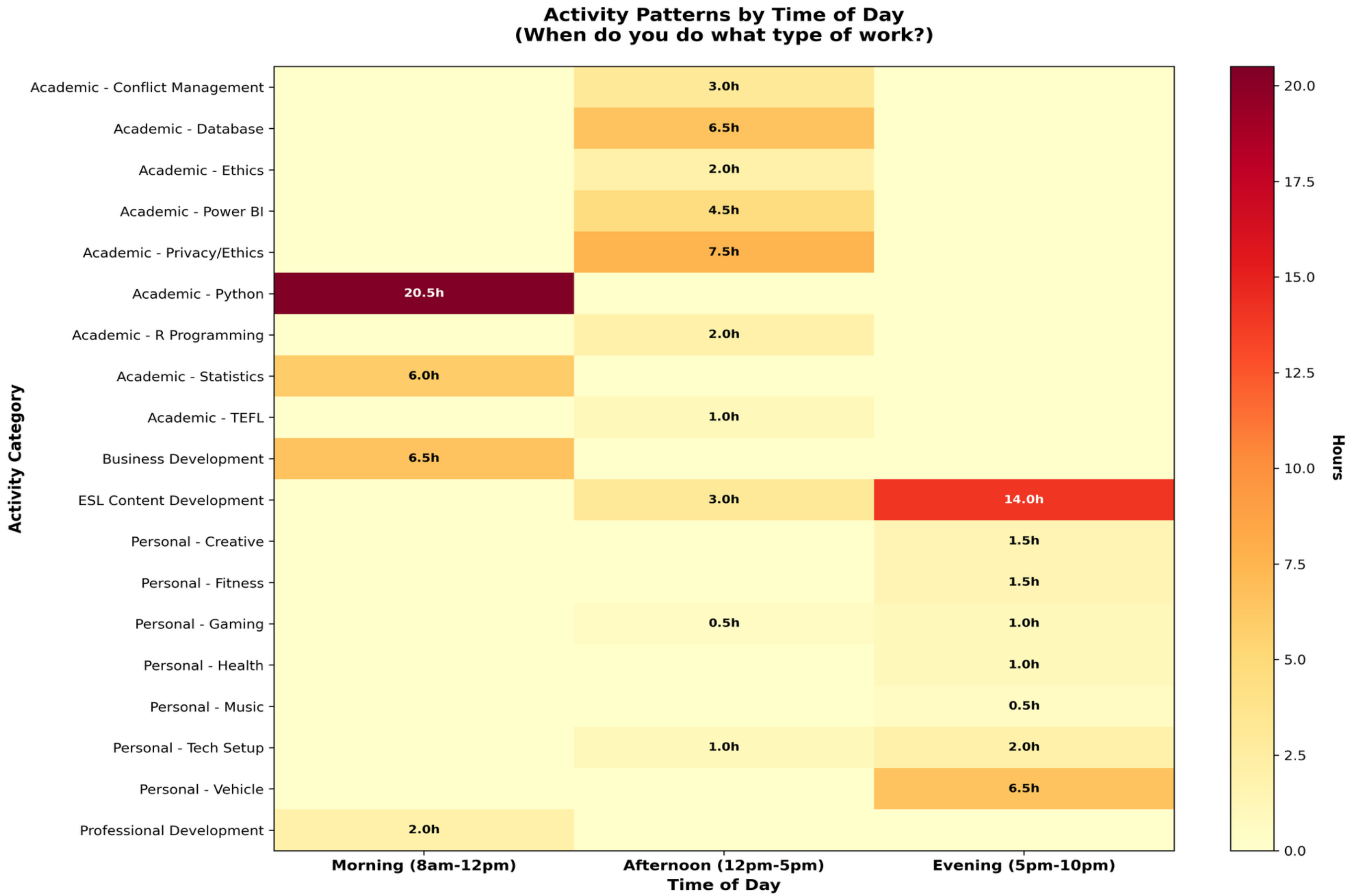


Task Flow Analysis:
- 91 tasks initiated
- 78 tasks completed
- 85.7% completion rate
- 13 tasks in progress (14.3%)

Work-In-Progress Metrics:
- Average daily WIP: 0.6 tasks
- Maintained low WIP throughout period

Completion Pattern:
- Consistent daily closure rate
- Most sessions: same-day completion
- Complex projects: 2-3 day completion

Finding 5: Work Type Naturally Aligns with Time Period



Analysis

Interpreting Performance Patterns

Pattern Analysis: Focus vs. Volume Capacity

1. Morning Focus Advantage
   - Energy (focus capacity): 7.9/10 - highest cognitive capacity
   - Handles complex work: avg complexity 9.3/10
   - Productivity for complex work: 124% higher than evening
   - Optimal for: Python, Database, Strategic Analysis

2. Evening Volume Advantage
   - Output capacity: 7.5/10 - highest task throughput
   - Completed 38 tasks vs 20 in morning
   - Lower complexity work: avg 4.8/10
   - Optimal for: Content creation, Admin, Quick tasks

3. Task Rotation & Rest Strategy
   - Alternating work types prevents fatigue
   - Strategic 20-minute rest intervals utilized
   - Low WIP (0.6 tasks/day) enables focus
   - 85.7% completion rate demonstrates discipline

Productivity Metrics: Complex Work Performance

0.731/hr

Morning Productivity

0.592/hr

Afternoon Productivity

0.326/hr

Evening Productivity

+124%

Morning Advantage

Improvement Opportunity: Optimize Time Allocation

Current Morning Allocation:

- 35 hours over 6 weeks (37.2%)
- Highest focus capacity window underutilized for complex tasks

Observed Misalignment:

- Complex analytical work sometimes scheduled in evening
- High-value tasks during lower-capacity periods
- Simple tasks occupying prime morning hours

Optimization Potential:

- Increase morning allocation: 35 → 45+ hours
- Reserve peak cognitive capacity for priority tasks
- Move high-volume simple tasks to evening

Expected Impact:

- Leverage 124% morning productivity advantage
- Maximize output quality on critical deliverables

Optimization

Data-Based Schedule Design

Proposed Schedule: Capacity-Task Alignment

Morning Block (8am-12pm)

Focus Capacity Zone

- Energy: 7.9/10 (highest cognitive capacity)

- Schedule: Complex analytical work
  - Python, Database development
  - Statistical analysis, Strategic planning

- Rationale: Quality (8.6) × Complexity (9.3) = Peak performance

Afternoon Block (12pm-5pm)

Implementation Zone

- Energy: 6.3/10 (moderate focus capacity)

- Schedule: Structured technical tasks
  - Power BI, SQL queries
  - Hands-on project work

- Rationale: Execute plans made in morning

Evening Block (5pm-10pm)

Volume Capacity Zone

- Energy: 3.7/10 (focus) BUT 7.5/10 (volume)

- Schedule: High-throughput activities
  - Content creation, Lesson plans
  - Administrative tasks
  - Quick wins

- Rationale: Leverage 38-task evening throughput advantage

Strategy: Match task complexity to time-of-day capacity for optimal performance

Implementation: Phased Schedule Transition

Phase 1: Immediate Reallocation (Week 1-2)

- Block 8am-12pm for complex analytical work only
- Move routine tasks to afternoon/evening slots
- Track energy and quality metrics daily

Phase 2: Refinement & Adjustment (Week 3-4)

- Monitor productivity scores using same formulas
- Adjust time block boundaries based on performance data
- Implement 20-min rest intervals as needed

Phase 3: Validation & Measurement (Week 5-6)

- Compare before/after productivity scores
- Calculate actual improvement percentage
- Document quality improvements on complex deliverables

Projected Performance Improvements

1. Productivity Score Increase

- Current: 0.548/hr
- Target: 0.620+/hr
- 13.1% improvement
- Method: Shift 15+ hours complex work to morning

2. Quality Output on Critical Tasks

- Leverage 124% morning advantage for priority work
- Maintain 8.6/10 quality on analytical deliverables
- Reduce quality variance across time periods

3. Task Management Efficiency

- Sustain 85%+ completion rate
- Keep WIP below 1.0 task/day average
- Improve same-day closure on complex items

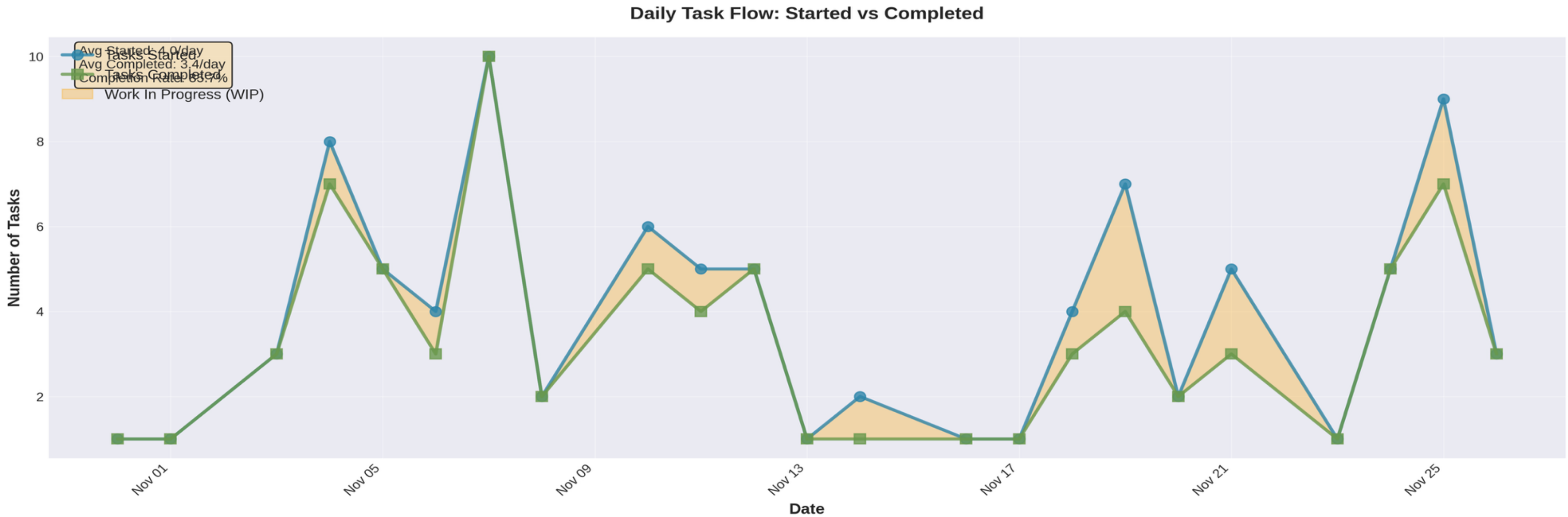
4. Energy Management

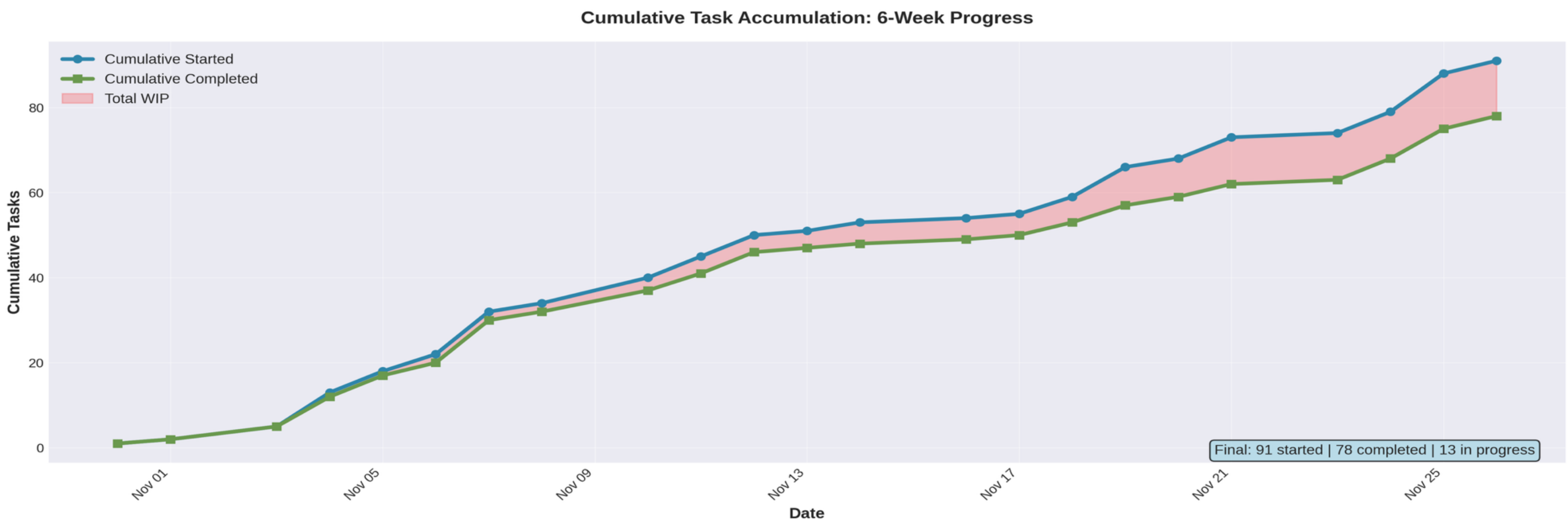
- Strategic task rotation prevents fatigue
- 20-minute rest intervals as needed
- Match task complexity to available cognitive capacity

Performance Data

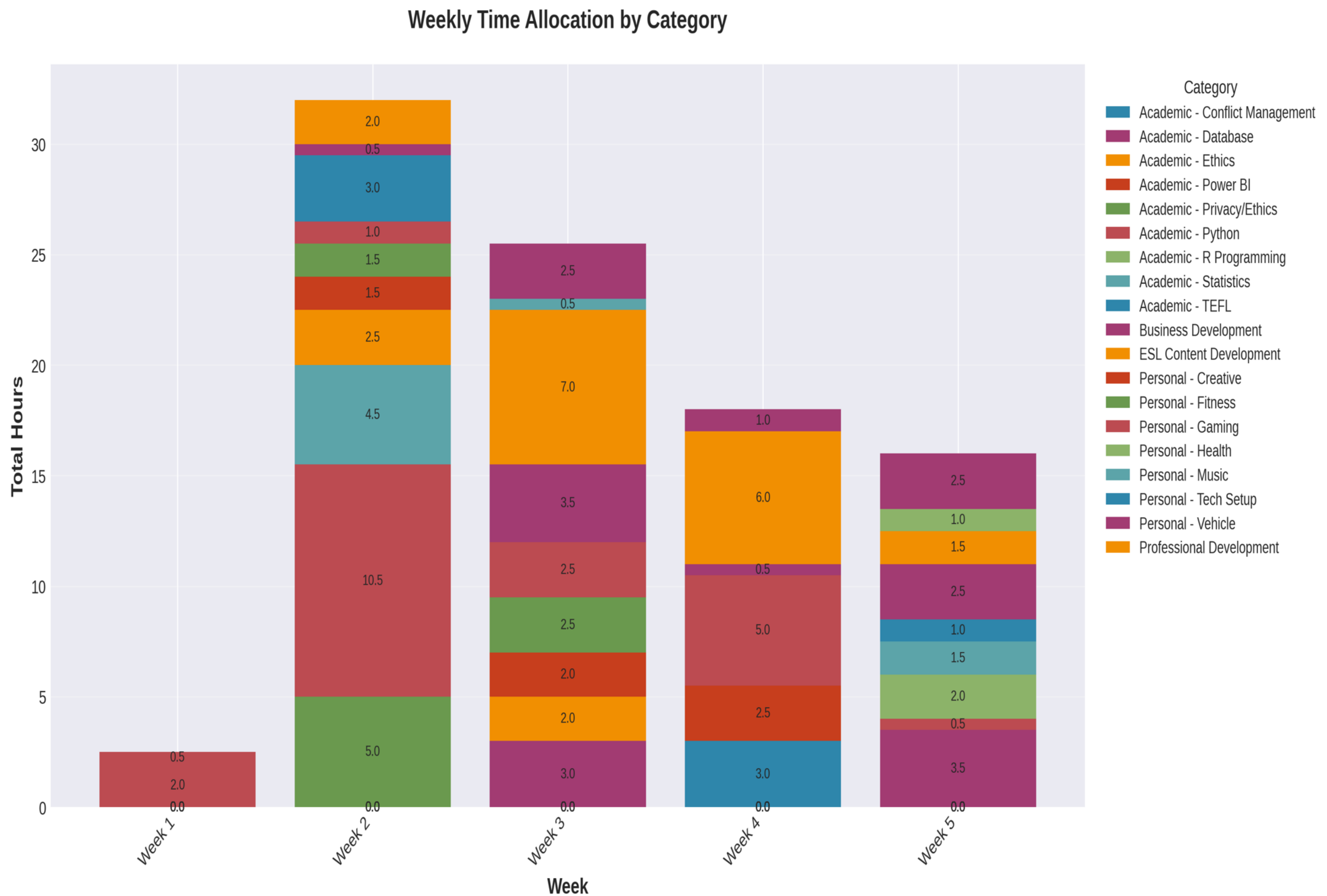
6-Week Documented Results

Task Completion: Sustained Performance Pattern





Weekly Performance Distribution



Measured Outcomes: Baseline Performance

Performance Metrics (6 Weeks):

✓ 94 hours documented work

✓ 57 work sessions measured

✓ 78 tasks completed
(85.7% completion rate)

✓ 8.3/10 avg output quality

✓ 5.9/10 avg energy (objective)

✓ 0.6 tasks WIP per day

Skill Development:

- Python/Data Analysis:
  20.5 hrs → 8.7/10 quality
- Database Management:
  6.5 hrs → 8.0/10 quality
- Business Intelligence:
  4.5 hrs → 8.0/10 quality
- Data Privacy/Ethics:
  7.5 hrs → 9.3/10 quality

Capacity Distribution:

- Morning: 7.9/10 focus capacity
  (complex work)
- Evening: 7.5/10 volume capacity
  (simple tasks)

Conclusions & Recommendations

1. Cognitive Capacity Varies by Time

- Morning: 7.9/10 focus capacity (124% productivity advantage)
- Evening: 7.5/10 volume capacity (best for high-throughput work)

Recommendation: Match task complexity to capacity profile

2. Objective Energy Validates Design

- Energy = (Quality × Complexity) ÷ 10
- Captures cognitive deployment, not subjective feeling

Recommendation: Use formula to evaluate schedule effectiveness

3. Task Completion Discipline

- Low WIP (0.6 tasks/day) correlates with 85.7% completion

Recommendation: Limit concurrent tasks, focus on closure

4. Work Variety Prevents Fatigue

- Task rotation maintains performance
- Strategic 20-minute intervals support sustained output

Recommendation: Build alternation and rest into schedule

Next Step: Implement optimized schedule and measure 13.1% improvement

Progressive Checkpoints: Data-Driven Adjustments

Checkpoint	Morning Hours	Key Metrics	Observation & Action
Week 2 (Nov 1-14)	21.5 hrs (39.1%)	Quality: 8.2 Completion: 92%	Pattern emerging: Morning shows highest energy (8.2) vs evening (3.5) Action: Schedule more Python/Database work in morning slots
Week 4 (Nov 1-21)	35.0 hrs (37.2%)	Quality: 8.2 Completion: 86%	Pattern confirmed: Morning optimal for complexity 9-10 tasks Action: Maintain morning priority for analytical work
Week 6 (Nov 1-26)	35.0 hrs (37.2%)	Quality: 8.2 Completion: 86%	Validated: 124% productivity advantage in morning for complex work Next Phase: Increase to 45+ morning hours per 6-week cycle

Key Insight: Progressive tracking revealed consistent morning advantage. Evening completes more tasks (38 vs 20) but morning delivers higher cognitive value (complexity 9.3 vs 4.8). Data validates capacity-task matching strategy.

Trends & Optimization Strategy

Metrics Progression

Morning Allocation:

- Week 2: 21.5 hrs (39%)
- Week 4: 35.0 hrs (37%)
- Week 6: 35.0 hrs (37%)

Quality Consistency:

- Week 2: 8.18/10
- Week 4: 8.19/10
- Week 6: 8.19/10

(Stable high quality maintained)

Completion Rate:

- Week 2: 92.2%
- Week 4: 85.7%
- Week 6: 85.7%

(Normalized as workload increased)

Optimization Roadmap

Phase 1 (Weeks 7-8):

Increase morning to 40 hours
Target: +5 hours complex work
Monitor: Quality & completion

Phase 2 (Weeks 9-10):

Increase morning to 45 hours
Reserve evening for volume tasks
Validate: Productivity gain ≥10%

Phase 3 (Weeks 11-12):

Stabilize at optimal allocation
Measure 13.1% improvement
Document refined schedule

Expected Outcome: By shifting 15 hours of complex analytical work to morning peak capacity periods, overall productivity should increase from 0.548 to 0.620 per hour. This represents a measurable 13.1% efficiency gain while maintaining quality standards and completion rates.